

Types of programming language

There are lots of different ways to group programming languages, and most languages fall into more than one group. A useful way of grouping is according to the features a language has.

Styles of programming language

Different styles of programming are sometimes called paradigms. They represent varied ways of thinking about computation. Some styles of programming are better at solving particular problems than others. Sometimes, there's no obvious approach and programmers will simply choose the language they are most comfortable with.

Imperative

The imperative style of programming is best described as a recipe or a knitting pattern. It is a series of commands that are executed one after the other. The recipe changes the state, or condition, of the ingredients from uncooked to cooked. The state of a computer is the data stored in its memory. When it runs a program, the commands in the program change this state. Imperative languages include variables, which hold data, and control structures, such as loops and conditional branches.

Visual

The first style of programming that children encounter is often visual programming. This describes languages where the programmer fits together blocks that represent instructions. Many visual languages are designed as educational tools. They allow children, or other new programmers, to become familiar with programming concepts without needing to type in commands. This allows them to focus on solving the problem without having to worry about programming errors.

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